

2

Classical propagation

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The propagation of light through an optical medium was discussed in general terms in Sections 1.1–1.3 of Chapter 1. We saw there that the propagation is characterized by two parameters, namely the refractive index and the absorption coefficient. In this chapter we shall investigate the classical theory of optical propagation, in which the light is treated as electromagnetic waves and the atoms or molecules are modelled as classical dipole oscillators. We shall see that this model gives a good general overview of the optical properties, and enables us to calculate the frequency dependence of the complex dielectric constant. This gives us the frequency dependence of the absorption coefficient and refractive index, and hence enables us to explain the phenomenon of dispersion. We shall also see that the model provides the framework for describing the effects due to optical anisotropy such as birefringence.

The treatment given here presupposes a working knowledge of the electromagnetic properties of dielectrics. A summary of the main results that we shall use is given in Appendix A. The model will be revisited in subsequent chapters when we consider the optical properties of free electrons in Chapter 7, and when we discuss lattice vibrations in Chapter 10. The model is also the starting point for the treatment of nonlinear optical effects in Chapter 11.

2.1 Propagation of light in a dense optical medium

The classical model of light propagation was developed at the end of the nineteenth century following Maxwell's theory of electromagnetic waves and the introduction of the concept of the dipole oscillator. In this section we shall give a qualitative discussion of the physical assumptions of this model, leaving the quantitative calculations to the next section.

The model assumes that there are several different types of oscillators within a medium, each with its own characteristic resonant frequency. In an insulator or semiconductor, the most important contribution at optical frequencies is from the oscillations of the **bound electrons** within the atoms, and so we begin this section by considering atomic oscillators. We then go on to introduce the idea of **vibrational oscillators**, which resonate at lower frequencies in the infrared spectral region, and finally mention **free electron oscillators**, which are responsible for the principal optical properties of metals.

2.1.1 Atomic oscillators

The concept of the dipole oscillator was introduced soon after Maxwell's electromagnetic theory. It was shown theoretically that an oscillating electric dipole would emit electromagnetic waves, and this was confirmed in 1887 when Heinrich Hertz succeeded in generating and detecting radio waves in the laboratory. He used an oscillatory discharge across a spark gap as the source and a wire loop as the aerial of the detector. This was an elegant confirmation of the validity of Maxwell's electromagnetic theory, and the beginning of radio telecommunications.

The idea of considering atoms as oscillating dipoles was originally proposed by Henrick Antoon Lorentz in 1878, thus preceding Hertz's demonstration by several years. It was known that atoms emit and absorb at discrete frequencies, and Lorentz's model provided a simple explanation for these observations in terms of the newly discovered electromagnetic theories.

The oscillator model of the atom is illustrated schematically in Fig. 2.1. It is assumed that the electron is held in a stable orbit with respect to the nucleus, and the spring represents the restoring force for small displacements from the equilibrium. The negatively charged electron and the positively charged nucleus form an electric dipole with a magnitude proportional to their separation. Lorentz, of course, could not have known about electrons and nuclei, because they were not discovered until 1897 and 1911 by J.J. Thomson and Ernest Rutherford respectively. Lorentz simply postulated the existence of dipoles without knowing their origin.

The natural resonant frequency ω_0 of the atomic dipoles is determined by their mass and the magnitude of the restoring force experienced for small displacements. The appropriate mass is the **reduced mass** given by:

$$\frac{1}{\mu} = \frac{1}{m_0} + \frac{1}{m_N}, \quad (2.1)$$

where m_0 and m_N are the masses of the electron and nucleus respectively. Since $m_N \gg m_0$, we may safely take $\mu \approx m_0$ here. The restoring force is quantified in terms of a spring constant K_s , which is chosen so that ω_0 coincides with one of the natural frequencies of the atoms (see Exercise 2.1):

$$\omega_0 = \sqrt{\frac{K_s}{\mu}}. \quad (2.2)$$

We have to suppose that there are several dipoles within every atom, to account for the fact that a given atom has many transition frequencies. These are known from the absorption and emission spectra, and the frequencies occur in the near-infrared, visible, and ultraviolet spectral regions (10^{14} – 10^{15} Hz).

We can understand the connection between the atomic dipoles and the emission spectra by considering the oscillations of the dipole shown in Fig. 2.2. An electric dipole consists of a positive charge $+q$ at position $\mathbf{r}_+$ and a negative charge $-q$ at $\mathbf{r}_-$. The electric-dipole moment is defined



Fig. 2.1 Classical model of the bound electrons in an atom. The electrons are represented by the open circles, while the black circle at the centre of the atom represents the nucleus. The electrons are held to the heavy nucleus by springs which represent the restoring forces due to the binding between them. Each atom has a series of characteristic resonant frequencies which we now know to correspond with the quantized transition energies.

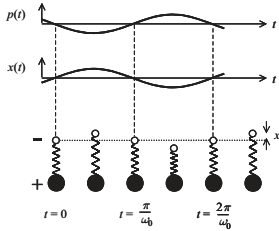


Fig. 2.2 Oscillations of a classical dipole consisting of a heavy positive charge and a light negative charge bound together by a spring. $x(t)$ is the time-dependent displacement of the negative charge from its equilibrium position. The natural vibrations of the dipole about the equilibrium position at angular frequency ω_0 generate a time-dependent dipole moment $p(t)$ as indicated in the top part of the figure.

as

$$\mathbf{p} = q(\mathbf{r}_+ - \mathbf{r}_-). \quad (2.3)$$

Hence the positive nucleus and negative electron form a dipole with magnitude equal to $e|\mathbf{r}_+ - \mathbf{r}_-|$.

During the oscillations of the atomic dipole, the nucleus remains more or less stationary due to its heavy mass, while the electron oscillates backwards and forwards at angular frequency ω_0 . Hence the oscillations produce a time-varying dipole in addition to any permanent dipole the atom might have. The magnitude of the time-varying dipole is given by:

$$p(t) = -ex(t), \quad (2.4)$$

where $x(t)$ is the time-varying displacement of the electron from its equilibrium position. This connection between the electron displacement and the time-dependent atomic dipole is illustrated in the top half of Fig. 2.2. The oscillating dipole behaves like a tiny aerial and radiates electromagnetic waves at angular frequency ω_0 , in accordance with the theory of classical Hertzian dipoles. Hence the atom is expected to radiate light at its resonant frequency whenever sufficient energy is imparted to excite the oscillations.

We can also use the dipole model to understand how the atom interacts with an external electromagnetic wave at angular frequency ω . The AC electric field exerts forces on the electron and the nucleus and drives oscillations of the system at frequency ω . If ω coincides with one of the natural frequencies of the atom, then we have a resonance phenomenon. This induces very large amplitude oscillations, and transfers energy from the external wave to the atom. The atom can therefore absorb energy from the light wave if $\omega = \omega_0$. The absorption strength is characterized by the absorption coefficient α , and the intensity of the wave will decay exponentially according to Beer's law (eqn 1.4).

We assume here that the forces exerted by the electric fields are very small compared to the binding forces that hold the electrons to the nucleus. This approximation may not be valid if we are using a very powerful laser beam to excite the medium. If this were the case, then we would be working in the regime of nonlinear optics. These effects are considered in Chapter 11.

We now know from quantum theory that what actually happens during absorption is that the atom jumps to an excited state by absorbing a photon. This can only occur if $\hbar\omega = E_2 - E_1$, where E_1 and E_2 are the quantized energies of the initial and final states. Once it has been excited, the atom can return to the ground state by a series of radiationless transitions, in which case the energy from the absorbed photon is ultimately converted into heat. Alternatively, it can luminesce by re-emitting a photon at some later time. The re-radiated photons are incoherent with each other and are emitted in all directions rather than in the specific direction of the incoming wave. Hence there is a net decrease in the energy flow in the beam direction, which is equivalent to absorption.

If ω does not coincide with any of the resonant frequencies, then the atoms will not absorb, and the medium will be transparent. In this situation the light wave drives non-resonant oscillations of the atoms at its own frequency ω . The oscillations of the atoms follow those of the driving wave, but with a phase lag. This phase lag is a standard feature of forced oscillators and is caused by damping. (See Exercise 2.2.) The oscillating atoms all re-radiate instantaneously, but the phase lag acquired in the process accumulates through the medium and retards the propagation of the wave front. This implies that the propagation velocity is smaller than in free space. The reduction of the velocity in the medium is characterized by the refractive index defined in eqn 1.2.

The slowing of the wave due to the non-resonant interactions can be considered as a repeated scattering process. The scattering is both coherent and elastic, and each atom behaves like a Huygens point source. The scattered light interferes constructively in the forward direction, and destructively in all other directions, so that the direction of the beam is unchanged by the repetitive scattering process. However, each scattering event introduces a phase lag which causes a slowing of the propagation of the phase front through the medium.

2.1.2 Vibrational oscillators

An optical medium may contain other types of dipole oscillators in addition to those originating from the bound electrons within the atoms. If the medium is ionic, it will contain oppositely charged ions. Vibrations of these charged atoms from their equilibrium positions within the crystal lattice will produce an oscillating dipole moment, in exactly the same way as the oscillations of the electrons within the individual atoms that we considered above. Therefore, we must also consider the optical effects due to these vibrational oscillators when we consider the interaction of light with an ionic optical medium.

The optical effects of vibrational oscillators are well known in molecular physics. Figure 2.3 gives a schematic illustration of a classical polar molecule. This consists of two charged atoms bound together in a stable configuration, with the spring representing the molecular bond between them. The charged atoms can vibrate about their equilibrium

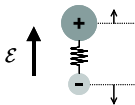


Fig. 2.3 Classical model of a polar molecule. The atoms are positively and negatively charged, and can vibrate about their equilibrium separation. These vibrations produce an oscillating electric dipole which will radiate electromagnetic waves at the resonant frequency. Alternatively, the molecule will interact with the electric field $\mathcal{E}$ of a light wave through the forces exerted on the charged atoms.

positions and induce an oscillating electric dipole in an analogous way to the bound electrons in the atoms. We see immediately from eqn 2.2 that the vibrations will occur at lower frequencies because the reduced mass is larger. The vibrations therefore occur at infrared frequencies with $\omega_0/2\pi \sim 10^{12}$ – 10^{13} Hz. These molecular vibrations are associated with strong absorption lines in the infrared spectral region.

The interaction between the vibrations of the molecule and the light wave occurs through the forces exerted on the atoms by the electric field. It is obvious that this can only happen if the atoms inside the molecule are charged. This is why we specified that the molecule was **polar** in the preceding paragraph. A polar molecule is one in which the charge cloud of the electrons that form the bond sits closer to one of the atoms than to the other. Ionic molecules like the alkali halides (e.g. Na^+Cl^-) clearly fall into this category, while purely covalent ones such as the elemental molecules (e.g. O_2) do not. Many other molecules fall somewhere between these two limits. Water (H_2O) is a well-known example. Oxygen has a greater electron affinity than hydrogen, and so the valence electrons in the O–H bond sit closer to the oxygen atoms. The two hydrogen atoms therefore possess a small positive charge which is balanced by a negative charge of twice the magnitude on the oxygen atom.

In a crystalline solid formed from the condensation of polar molecules, the atoms are arranged in an alternating sequence of positive and negative ions. The ions can vibrate about their equilibrium positions, and this produces oscillating dipole waves. These oscillations are associated with **lattice vibrations**, and they occur at frequencies in the infrared spectral region. We shall consider the optical properties related to the lattice vibrations in detail in Chapter 10. We shall see there that the light–matter interaction is associated with the excitation of **phonons**, which are quantized lattice waves. At this stage, we simply note that the lattice vibrations of a polar crystal give rise to strong optical effects in the infrared spectral region. These effects occur in addition to those due to the bound electrons of the atoms that comprise the crystal. In practice we can treat these two types of dipoles separately because the resonances occur at very different frequencies. Therefore the resonant effects of the bound electrons are negligible at the frequencies of the lattice vibrations, and vice versa. This point will be considered in more detail in Section 2.2.2.

2.1.3 Free electron oscillators

The electronic and vibrational dipoles considered above are both examples of bound oscillators. Metals and doped semiconductors, by contrast, contain significant numbers of **free electrons**. As the name implies, these are electrons that are not bound to any atoms, and therefore do not experience any restoring forces when they are displaced. This implies that the spring constant in eqn 2.2 is zero, and hence that the natural resonant frequency $\omega_0 = 0$.

The free electron model of metals is attributed to Paul Drude, and so the application of the dipole oscillator model to free electron systems is generally called the **Drude–Lorentz model**. The dipole oscillator model is perfectly valid, except that we must set $\omega_0 = 0$ throughout. The optical properties of free electron systems will be discussed in Chapter 7.

2.2 The dipole oscillator model

In the previous section we introduced the general assumptions of the dipole oscillator model. We now want to use the model to calculate the frequency dependence of the refractive index and absorption coefficient. This will provide a simple explanation for the dispersion of the refractive index in optical materials. It will also illustrate the general point, which will be developed further in Section 2.3, that the phenomena of absorption and refraction are related to each other.

2.2.1 The Lorentz oscillator

We consider the interaction between a light wave and an atom with a single resonant frequency ω_0 due to the bound electrons, as given by eqn 2.2. We model the displacement of the atomic dipoles as damped harmonic oscillators. The inclusion of damping is a consequence of the fact that the oscillating dipoles can lose their energy by collisional processes. In solids, this would typically occur through an interaction with a phonon which has been thermally excited in the crystal. As we shall see, the damping term has the effect of reducing the peak absorption coefficient and broadening the absorption line.

The electric field of the light wave induces forced oscillations of the atomic dipole through the driving forces exerted on the electrons. We make the reasonable assumption that the nuclear mass is much greater than the electron mass (i.e. $m_N \gg m_0$), so that we can ignore the motion of the nucleus. The displacement x of the electron is governed by an equation of motion of the form:

$$m_0 \frac{d^2x}{dt^2} + m_0 \gamma \frac{dx}{dt} + m_0 \omega_0^2 x = -e\mathcal{E}, \quad (2.5)$$

where γ is the damping rate, e is the magnitude of the electric charge of the electron, and $\mathcal{E}$ is the electric field of the light wave. The terms on the left-hand side represent the acceleration, the damping and the restoring force respectively. The damping is modelled by a frictional force which is proportional to the velocity and impedes the motion. The term on the right-hand side represents the driving force due to the AC electric field of the light wave.

We consider the interaction of the atom with a monochromatic light wave of angular frequency ω . The time dependence of the electric field is given by

$$\mathcal{E}(t) = \mathcal{E}_0 \cos(\omega t + \Phi) = \mathcal{E}_0 \operatorname{Re} \left(e^{-i(\omega t + \Phi)} \right), \quad (2.6)$$

We know from experimental observations that atoms must have many natural resonant frequencies to account for the multiplicity of lines in the absorption and emission spectra. However, the salient features of the physical behaviour are well illustrated by a singly resonant system, and the inclusion of multiple resonances complicates the discussion without adding much to the physical understanding at this stage. We therefore postpone the discussion of the effects of multiple resonances to Section 2.2.2.

In some conventions, the time dependence is written as $e^{j\omega t}$ instead of $e^{-i\omega t}$. This makes no physical difference. Consistency can be obtained by replacing $-i$ with $+j$ throughout.

Note that the phase factors Φ and Φ' in eqns 2.6 and 2.7 are not necessarily the same. In fact, the phase of the electrons will tend to lag behind the phase of the light. This is a well known property of forced oscillations: the vibrations occur at the same frequency as the driving force but lag behind due to the damping term. This phase lag is the origin of the slowing down of the light in the optical medium, and hence of the refractive index, as discussed above in Section 2.1

where $\mathcal{E}_0$ is the amplitude and Φ is the phase of the light. In order to keep consistency with the sign convention introduced later, we have chosen to take the negative frequency part of the complex exponential.

The AC electric field will drive oscillations at its own frequency ω . We therefore substitute eqn 2.6 into eqn 2.5 and look for solutions of the form:

$$x(t) = X_0 \Re \left(e^{-i(\omega t + \Phi')} \right), \quad (2.7)$$

where X_0 and Φ' are the amplitude and phase of the oscillations. We can incorporate the phase factors of eqns 2.6 and 2.7 into the amplitudes by allowing both $\mathcal{E}_0$ and X_0 to be complex numbers. We then substitute $\mathcal{E}(t) = \mathcal{E}_0 e^{-i\omega t}$ into eqn 2.5, and look for solutions of the form $x(t) = X_0 e^{-i\omega t}$. This gives:

$$-m_0 \omega^2 X_0 e^{-i\omega t} - i m_0 \gamma \omega X_0 e^{-i\omega t} + m_0 \omega_0^2 X_0 e^{-i\omega t} = -e \mathcal{E}_0 e^{-i\omega t}, \quad (2.8)$$

which implies that:

$$X_0 = \frac{-e \mathcal{E}_0 / m_0}{\omega_0^2 - \omega^2 - i \gamma \omega}. \quad (2.9)$$

The displacement of the electrons from their equilibrium positions produces a time-varying dipole moment $p(t)$, as shown in Fig. 2.2. The magnitude of the dipole is given by eqn 2.4. This gives a resonant contribution to the macroscopic polarization (dipole moment per unit volume) of the medium. If N is the number of atoms per unit volume, the resonant polarization is given by:

$$\begin{aligned} P_{\text{resonant}} &= N p \\ &= -N e x \\ &= \frac{N e^2}{m_0} \frac{1}{(\omega_0^2 - \omega^2 - i \gamma \omega)} \mathcal{E}. \end{aligned} \quad (2.10)$$

A quick inspection of eqn 2.10 shows that the magnitude of P_{resonant} is small unless the frequency is close to ω_0 . This is another general property of forced oscillations: the response is small unless the frequency is close to resonance with the natural frequency of the oscillator.

Equation 2.10 can be used to obtain the complex relative dielectric constant ϵ_r . The electric displacement $\mathbf{D}$ of the medium is related to the electric field $\mathcal{E}$ and polarization $\mathbf{P}$ through:

$$\mathbf{D} = \epsilon_0 \mathcal{E} + \mathbf{P}, \quad (2.11)$$

where the bold font indicates vector quantities (see eqn A.2 in Appendix A). We are interested in the optical response at frequencies close to ω_0 , and so we split the polarization into a non-resonant background term and the resonant term arising from the driven response of the oscillator. We therefore write:

$$\begin{aligned} \mathbf{D} &= \epsilon_0 \mathcal{E} + \mathbf{P}_{\text{background}} + \mathbf{P}_{\text{resonant}} \\ &= \epsilon_0 \mathcal{E} + \epsilon_0 \chi \mathcal{E} + \mathbf{P}_{\text{resonant}}. \end{aligned} \quad (2.12)$$

The electric susceptibility χ in eqn 2.12 accounts for all other contributions to the polarizability of the atoms. We shall discuss the physical meaning of the 'non-resonant polarization' in Section 2.2.2 below.

To simplify the mathematics, we assume that the material is isotropic, in which case the relative dielectric constant is defined through the relationship:

$$\mathbf{D} = \epsilon_0 \epsilon_r \mathbf{E}. \quad (2.13)$$

We then combine eqns 2.10–2.13 to obtain:

$$\epsilon_r(\omega) = 1 + \chi + \frac{Ne^2}{\epsilon_0 m_0} \frac{1}{(\omega_0^2 - \omega^2 - i\gamma\omega)}. \quad (2.14)$$

The treatment of non-isotropic materials only introduces unnecessary complications at this stage, and so the consideration of anisotropy will be postponed to Section 2.5.1.

This can be split into its real and imaginary parts according to eqn 1.21 to give:

$$\epsilon_1(\omega) = 1 + \chi + \frac{Ne^2}{\epsilon_0 m_0} \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2} \quad (2.15)$$

$$\epsilon_2(\omega) = \frac{Ne^2}{\epsilon_0 m_0} \frac{\gamma\omega}{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2}. \quad (2.16)$$

These formulae can be simplified further if we are working at frequencies close to resonance, where $\omega \approx \omega_0 \gg \gamma$. This allows us to approximate $(\omega_0^2 - \omega^2)$ by $2\omega_0\Delta\omega$, where $\Delta\omega = (\omega - \omega_0)$ is the detuning from ω_0 . We then notice that the low and high frequency limits of $\epsilon_r(\omega)$ are given by

$$\epsilon_r(0) \equiv \epsilon_{st} = 1 + \chi + \frac{Ne^2}{\epsilon_0 m_0 \omega_0^2}, \quad (2.17)$$

and

$$\epsilon_r(\infty) \equiv \epsilon_\infty = 1 + \chi, \quad (2.18)$$

respectively. The subscript on ϵ_{st} stands for ‘static’, since it represents the dielectric response to static (i.e. low-frequency) electric fields. With this notation we find that:

$$(\epsilon_{st} - \epsilon_\infty) = \frac{Ne^2}{\epsilon_0 m_0 \omega_0^2}. \quad (2.19)$$

We finally rewrite eqns 2.15 and 2.16 in the following form valid at frequencies close to resonance:

$$\epsilon_1(\Delta\omega) = \epsilon_\infty - (\epsilon_{st} - \epsilon_\infty) \frac{2\omega_0\Delta\omega}{4(\Delta\omega)^2 + \gamma^2}, \quad (2.20)$$

$$\epsilon_2(\Delta\omega) = (\epsilon_{st} - \epsilon_\infty) \frac{\gamma\omega_0}{4(\Delta\omega)^2 + \gamma^2}. \quad (2.21)$$

These equations describe a sharp atomic absorption line centred at ω_0 with full width at half maximum equal to γ .

Figure 2.4 shows the frequency dependence of ϵ_1 and ϵ_2 predicted by eqns 2.20–2.21 for an oscillator with $\omega_0 = 10^{14}$ rad/s, $\gamma = 5 \times 10^{12}$ s⁻¹, $\epsilon_{st} = 12.1$ and $\epsilon_\infty = 10$. We see that ϵ_2 is a strongly peaked function of ω with a maximum value at ω_0 and a full width at half maximum equal to γ . The frequency dependence of ϵ_1 is more complicated. As we approach ω_0 from below, ϵ_1 gradually rises from the low-frequency value of ϵ_{st} , and reaches a peak at $\omega_0 - \gamma/2$. (See Example 2.1 below.) It

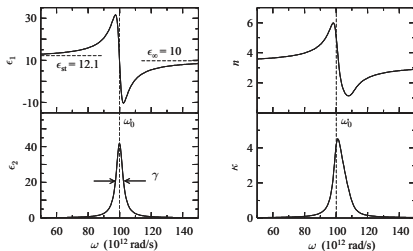


Fig. 2.4 Frequency dependence of the real and imaginary parts of the complex dielectric constant of a dipole oscillator at frequencies close to resonance. The graphs are calculated for an oscillator with $\omega_0 = 10^{14}$ rad/s, $\gamma = 5 \times 10^{12}$ s $^{-1}$, $\epsilon_{st} = 12.1$, 0 and $\epsilon_\infty = 10$. Also shown is the real and imaginary part of the refractive index calculated from the dielectric constant.

then falls sharply, passing through a minimum at $\omega_0 + \gamma/2$ before rising again to the high-frequency limit of ϵ_∞ . Note that the frequency scale over which these effects occur is determined by γ for both ϵ_1 and ϵ_2 . This shows that the damping of the oscillator causes line broadening. The frequency dependence determined of ϵ_1 and ϵ_2 shown in Fig. 2.4 is called **Lorentzian** after the originator of the dipole model.

In an experiment we actually measure the refractive index n and the absorption coefficient α . The measurement of α then determines the extinction coefficient κ through eqn 1.19. Figure 2.4 shows the values of n and κ calculated from ϵ_1 and ϵ_2 using eqns 1.25 and 1.26. We see that n approximately follows the frequency dependence of $\sqrt{\epsilon_1(\omega)}$, while κ more or less follows $\epsilon_2(\omega)$. The correspondence $n \leftrightarrow \sqrt{\epsilon_1}$ and $\kappa \leftrightarrow \epsilon_2$ would be exact if κ were much smaller than n (cf. eqns 1.27 and 1.28). This is what generally happens in gases in which the low density of atoms makes the total absorption small. In the example shown in Fig. 2.4, the correspondence is only approximate because the absorption is very strong near ω_0 , so that we cannot always assume $n \gg \kappa$. Nevertheless, the basic behaviour shows that the absorption peaks at a frequency very close to ω_0 and has a width of about γ , while the refractive index shows positive and negative excursions below and above ω_0 . This is the typical behaviour expected of an atomic absorption line.

One important aspect of a resonant Lorentzian dipole is that the effect on the refractive index occurs over a much larger frequency range than on the absorption. This point is clearly illustrated in the graphs shown in Fig. 2.4. The absorption is a strongly-peaked function of ω and falls off as $(\Delta\omega)^{-2}$ as we tune away from resonance. Thus there is no significant

absorption if we tune sufficiently far from resonance. On the other hand, the frequency dependence of the refractive index varies more slowly as $|\Delta\omega|^{-1}$ for large $|\Delta\omega|$. This follows from eqn 2.20 with the approximation $n = \sqrt{\epsilon_1}$, which is valid for large $|\Delta\omega|$ when ϵ_2 is very small. This means that there can still be a significant contribution to the refractive index even when the light frequency is far from resonance with the absorption frequency.

Example 2.1

The full width at half maximum of an atomic absorption line at 589.0 nm is 100 MHz. A beam of light passes through a gas with an atomic density of $1 \times 10^{17} \text{ m}^{-3}$. Calculate:

- The peak absorption coefficient due to this absorption line.
- The frequency at which the resonant contribution to the refractive index is at a maximum.
- The peak value of the resonant contribution to the refractive index.

Solution

(a) We are dealing with a gas with a low density of atoms, and so the approximations given in eqns 1.27 and 1.28 will be valid. This means that the absorption will directly follow the frequency dependence of $\epsilon_2(\omega)$, and the peak absorption will occur precisely at the line centre. The peak extinction coefficient can be worked out from eqns 2.16 and 1.28. This gives:

$$\kappa(\omega_0) = \frac{\epsilon_2(\omega_0)}{2n} = \frac{Ne^2}{2n\epsilon_0 m_0} \frac{1}{\gamma\omega_0}.$$

We do not know what n is, but because we are dealing with a gas, it will only be very slightly different from unity. This point is confirmed in part (c) of the example. We therefore take $n = 1$ here, and insert $N = 1 \times 10^{17} \text{ m}^{-3}$, $\gamma = 2\pi \times 10^8 \text{ s}^{-1}$ and $\omega_0 = 2\pi c/\lambda = 3.20 \times 10^{15} \text{ rad/s}$, to find that $\kappa(\omega_0) = 7.90 \times 10^{-5}$. This confirms that $n \gg \kappa$, and hence that it is valid to use eqn 1.28. We then work out the absorption coefficient from eqn 1.19, which gives:

$$\alpha_{\max} \equiv \alpha(\omega_0) = \frac{4\pi\kappa(\omega_0)}{\lambda} = 1.7 \times 10^3 \text{ m}^{-1}.$$

(b) We know from Fig. 2.4 that there will be a peak in the refractive index just below ω_0 . Equation 1.27 tells us that $n(\omega) = \sqrt{\epsilon_1(\omega)}$, and hence that the local maximum of n will occur at the same frequency as the maximum in ϵ_1 . Since the peak occurs near ω_0 , it will be valid to use eqn 2.20. The local maximum occurs when:

$$\frac{d\epsilon_1(\omega)}{d\omega} \equiv \frac{d\epsilon_1(\Delta\omega)}{d\Delta\omega} \propto \frac{4(\Delta\omega)^2 - \gamma^2}{[4(\Delta\omega)^2 + \gamma^2]^2} = 0.$$

The wavelength and linewidth of the transition considered in this example correspond to the strongest hyperfine component of the sodium D₂ line. The true absorption coefficient of this transition is smaller than the value calculated here by about a factor of 3. This discrepancy occurs because we are erroneously assuming that the oscillator strength of the transition is equal to unity, which it is not. The concept of the oscillator strength is introduced in Section 2.2.2.

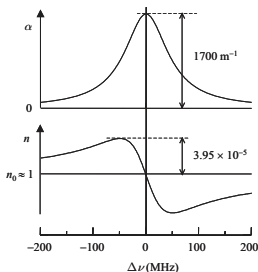


Fig. 2.5 Absorption coefficient and refractive index of the atomic gas considered in Example 2.1. n_0 represents the off-resonant refractive index, which is approximately equal to unity.

This gives $\Delta\omega = \pm\gamma/2$. We see from Fig. 2.4 that $\Delta\omega = -\gamma/2$ corresponds to the local maximum, while $\Delta\omega = +\gamma/2$ corresponds to the local minimum. Therefore the peak in the refractive index occurs 50 MHz below the line centre.

(c) From part (b) we know that the local maximum in the refractive index occurs when $\Delta\omega = -\gamma/2$. We see from eqns 1.27 and 2.20 that the refractive index at this frequency is given by:

$$n_{\max} = \sqrt{\epsilon_1} = \left(\epsilon_\infty + \frac{Ne^2}{2\epsilon_0 m_0 \omega_0 \gamma} \right)^{1/2} = n_0 \left(1 + \frac{7.90 \times 10^{-5}}{n_0^2} \right)^{1/2},$$

where $n_0 = \sqrt{\epsilon_\infty}$ is the off-resonant refractive index. We are dealing with a low-density gas, and so it is justified to take $n_0 \approx 1$ here. This implies that the peak value of the resonant contribution to the refractive index is 3.95×10^{-5} .

The full frequency dependence of the absorption and refractive index near this absorption line is plotted in Fig. 2.5.

2.2.2 Multiple resonances

In general, an optical medium will have many characteristic resonant frequencies. We already discussed in Section 2.1 how we expect to observe separate resonances due to the lattice vibrations and to the oscillations of the bound electrons within the atoms. Furthermore, a particular medium may have many resonances of each type. We can treat these multiple resonances without difficulty in our model provided they occur at distinct frequencies.

In writing eqn 2.12 we split the polarization of the medium into a resonant part and a non-resonant part. We then discussed the resonant part in detail, without specifying very accurately what we meant by the 'non-resonant' term. We simply stated that $\mathbf{P}$ was proportional to $\mathcal{E}$ through the susceptibility χ . In reality, the non-resonant polarization of the medium must originate from the polarizability of the atoms in exactly the same way as the resonant part. Equation 2.19 tells us that the dielectric constant decreases each time we go through an absorption line. The contributions that enter the background electric susceptibility χ in eqn 2.12 thus arise from the polarization due to all the other oscillators at higher frequencies.

We can understand this point better by making it more quantitative. The contribution to the polarization of a particular oscillator is given by eqn 2.10. In a medium with many electronic oscillators of different frequencies, the total polarization will therefore be given by

$$\mathbf{P} = \left(\frac{Ne^2}{m_0} \sum_j \frac{1}{(\omega_j^2 - \omega^2 - i\gamma_j\omega)} \right) \mathcal{E}, \quad (2.22)$$

where ω_j and γ_j are the angular frequency and damping coefficient of a particular resonance line. We then substitute this into eqn 2.11, and recall the definition of ϵ_r given in eqn 2.13, to obtain:

$$\epsilon_r(\omega) = 1 + \frac{Ne^2}{\epsilon_0 m_0} \sum_j \frac{1}{(\omega_j^2 - \omega^2 - i\gamma_j\omega)}. \quad (2.23)$$

This equation takes account of all the transitions in the medium and can be used to calculate the full frequency dependence of the dielectric constant.

The refractive index and absorption coefficient calculated from eqn 2.23 are plotted against frequency in Fig. 2.6. The figure has been calculated for a hypothetical solid with three well-separated resonances with ω_j equal to 4×10^{13} rad/s, 4×10^{15} rad/s, and 1×10^{17} rad/s respectively. The width of each absorption line has been set to 10% of the centre frequency by appropriate choice of the damping coefficient. The resonance in the infrared is included to represent the vibrational absorption. In a real solid, we would have to adapt the model appropriately to account for the different reduced mass and effective charge of the vibrational oscillator.

We can understand this figure by starting at the highest frequencies and gradually working our way down to the lower frequencies. At the very highest frequencies, the electrons are too sluggish to respond to the driving field. The medium therefore has no polarization, and the dielectric constant is unity. As we reduce the frequency, we first run into the transitions of the inner electrons in the X-ray/vacuum-ultraviolet spectral region, and then the transitions of the outer electrons in the ultraviolet and visible. We then have a region with no transitions until we finally reach the vibrational frequencies in the infrared. Each time we

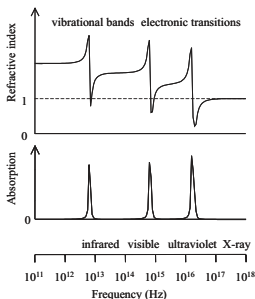


Fig. 2.6 Schematic diagram of the frequency dependence of the refractive index and absorption of a hypothetical solid from the infrared to the X-ray spectral region. The solid is assumed to have three resonant frequencies with $\omega_j = 4 \times 10^{13}$ rad/s, 4×10^{15} rad/s, and 1×10^{17} rad/s respectively. The width of each absorption line has been set to 10% of the centre frequency by appropriate choice of γ_j .

go through one of these resonances, we see the characteristic frequency dependence of the Lorentz oscillator, with a peak in the absorption spectrum and a ‘wiggle’ in the refractive index. In between the resonances the medium is transparent: the absorption coefficient is zero and the refractive index is almost constant.

The value of the refractive index in the transparent regions gradually increases as we go through more and more resonance lines on decreasing the frequency. This increase of the refractive index is caused by the fact that $\epsilon_{\text{st}} > \epsilon_{\infty}$ (cf. eqn 2.19), which implies that n is larger below an absorption line than above it. By reference to Fig. 2.6, we now see that we have to understand ‘static’ and ‘ ∞ ’ as relative to a particular resonance. The variation of n with frequency due to the resonances is the origin of the dispersion found in optical materials even when they are transparent. This point will be discussed further in Section 2.4.

The dipole oscillator model predicts that each oscillator contributes a term given by eqn 2.10. This leads to a series of absorption lines of the same strength. However, experimental data shows that the absorption strength actually varies considerably between different atomic transitions. With the benefit of hindsight, we know that this is caused by the variation of the quantum mechanical transition probability. (See Appendix B.) In classical physics, however, there is no explanation, and we just assign a phenomenological **oscillator strength** f_j to each transition, rewriting eqn 2.23 as:

An astute reader will have noticed that the peak absorption coefficient for the three transition lines shown in Fig. 2.6 decreases slightly with decreasing frequency. This happens because n is larger at the lower frequencies. The transitions all have the same peak ϵ_2 , but we can see from eqn 1.24 that κ must be slightly smaller if n is larger.

$$\epsilon_r(\omega) = 1 + \frac{Ne^2}{\epsilon_0 m_0} \sum_j \frac{f_j}{(\omega_{0j}^2 - \omega^2 - i\gamma_j\omega)}. \quad (2.24)$$

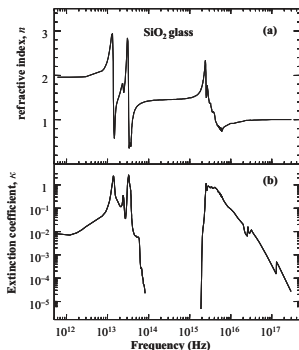


Fig. 2.7 (a) Refractive index and (b) extinction coefficient of fused silica (SiO_2) glass from the infrared to the x-ray spectral region. After Palik (1985).

It can be shown from quantum mechanics that we must have $\sum_j f_j = 1$ for each electron. Since the classical model predicts $f_j = 1$ for each oscillator, we then interpret this by saying that a particular electron is involved in several transitions at the same time, and the absorption strength is being divided between these transitions.

2.2.3 Comparison with experimental data

The schematic behaviour shown in Fig. 2.6 can be compared to experimental data on a typical optical material. Figure 2.7 shows the frequency dependence of the refractive index and extinction coefficient of fused silica (SiO_2) glass from the infrared to the X-ray spectral region. The general characteristics indicated by Fig. 2.6 are clearly observed, with strong absorption in the infrared and ultraviolet, and a broad region of low absorption in between. The data confirms that $n \gg \kappa$ except near the peaks of the absorption. This means that the approximation whereby we associate the frequency dependence of n with that of ϵ_1 , and that of κ with ϵ_2 (eqns 1.27 and 1.28), is valid at most frequencies.

The general behaviour shown in Fig. 2.7 is typical of optical materials which are transparent in the visible spectral region. We already noted in Sections 1.4.1 and 1.4.2 that the transmission range of colourless materials is determined by the electronic absorption in the ultraviolet and the vibrational absorption in the infrared. This is demonstrated by the

transmission data for sapphire shown in Fig. 1.4(a).

Fused silica is a glass, and hence does not have a regular crystal lattice. The infrared absorption is therefore caused by excitation of vibrational quanta in the SiO_2 molecules themselves. Two distinct peaks are observed at 1.4×10^{13} Hz (21 μm) and 3.3×10^{13} Hz (9.1 μm) respectively. These correspond to different vibrational modes of the molecule. The detailed modelling of these absorption bands by the oscillator model will be discussed in Chapter 10.

The ultraviolet absorption in silica is caused by interband electronic transitions. SiO_2 has a fundamental band gap of about 10 eV, and interband transitions are possible whenever the photon energy exceeds this value. Hence we observe an absorption threshold in the ultraviolet at 2×10^{15} Hz (150 nm). The interband absorption peaks at around 3×10^{15} Hz with an extremely high absorption coefficient of $\sim 10^8 \text{ m}^{-1}$, and then gradually falls off to higher frequency. Subsidiary peaks are observed at $\sim 3 \times 10^{16}$ Hz and 1.3×10^{17} Hz. These are caused by transitions of the inner core electrons of the silicon and oxygen atoms. The fact that the electronic absorption consists of a continuous band rather than a discrete line makes it hard to model accurately as a Lorentz oscillator. We shall discuss the quantum theory of the interband absorption in Chapter 3.

The refractive index of glass has resonances in the infrared and the ultraviolet which correspond respectively to the vibrational and interband absorption bands. In the far infrared region below the vibrational resonance, the refractive index is ~ 2 , while in the hard ultraviolet and X-ray region it approaches unity. In the transparency region between the vibrational and interband absorption, the refractive index has a value of ~ 1.5 . Closer inspection of Fig. 2.7 shows that the refractive index actually increases with frequency in this transparency region, rising from a value of 1.40 at 8×10^{13} Hz (3.5 μm) to 1.55 at 1.5×10^{15} Hz (200 nm). This dispersion originates from the low-frequency wings of the ultraviolet absorption and the high-frequency wings of the infrared absorption, and will be discussed in more detail in Section 2.4.

The data in Fig. 2.7 show that the refractive index falls below unity at a number of frequencies. This implies that the phase velocity of the light is greater than c , which might seem to imply a contradiction with relativity. However, this overlooks the fact that a signal must be transmitted as a wave packet rather than as a monochromatic wave. In a dispersive medium, a wave packet will propagate at the **group velocity** v_g given by:

$$v_g = \frac{d\omega}{dk}, \quad (2.25)$$

rather than at the phase velocity $v = \omega/k = c/n$. The relationship between v_g and v is:

$$v_g = v \left(1 + \frac{\omega}{n} \frac{dn}{d\omega} \right)^{-1} = v \left(1 - \frac{\lambda}{n} \frac{dn}{d\lambda} \right)^{-1}, \quad (2.26)$$

where λ is the vacuum wavelength of the light. The derivation of this

It is apparent from Fig. 2.6 that $dn/d\omega$ will be negative at some frequencies close to a resonance line. Equation 2.26 then implies that $v_g > v$, and so we could again run into a problem with relativity. However, the medium is highly absorbing in these frequency regions, and this means that the signal travels with yet another velocity called the signal velocity. This is always less than c .

result is left as an exercise to the reader. (See Exercise 2.7.) We shall see in Section 2.4 that $dn/d\omega$ is positive in most materials at optical frequencies. This then implies that v_g is always less than v , and if we were to try to transmit a signal in a spectral region where $v > c$, we would always find that v_g is less than c . The proof of this for a simple Lorentz oscillator is considered in Exercise 2.8.

2.2.4 Local field corrections

The calculation of the dielectric constant given in eqn 2.24 is valid in a rarefied gas with a low density of atoms. However, in a dense optical medium such as a solid, there is another factor that we must consider. The individual atomic dipoles respond to the **local field** that they experience. This may not necessarily be the same as the external field, because the dipoles themselves generate electric fields that will be felt by all the other dipoles. The actual local field experienced by an atom therefore takes the form:

$$\mathcal{E}_{\text{local}} = \mathcal{E} + \mathcal{E}_{\text{other dipoles}}, \quad (2.27)$$

where $\mathcal{E}$ and $\mathcal{E}_{\text{other dipoles}}$ represent the fields due to the external field and the other dipoles respectively. We should have been using $\mathcal{E}_{\text{local}}$ instead of $\mathcal{E}$ all along throughout the calculation given in Sections 2.2.1 and 2.2.2.

The calculation of the correction field due to the other dipoles in the medium is actually a rather complicated one. An approximate solution due to Lorentz can be derived if we assume that all the dipoles are parallel to the applied field and are arranged on a cubic lattice. The calculation works by separating the contribution from the nearby dipoles and that from the rest of the sample, as indicated in Fig. 2.8. The division is effected by an imaginary spherical surface with a radius large enough to make it sensible to average the material outside it. The problem is then reduced to summing the field of the dipoles inside the sphere at the one in the middle, and then calculating the effect of a uniformly polarized dielectric outside the sphere. The final result is:

$$\mathcal{E}_{\text{other dipoles}} = \frac{\mathbf{P}}{3\epsilon_0}, \quad (2.28)$$

where $\mathbf{P}$ is the polarization of the dielectric outside the sphere. The derivation of this result is the subject of Exercise 2.9. By using the result of eqn 2.28 in eqn 2.27 we find that:

$$\mathcal{E}_{\text{local}} = \mathcal{E} + \frac{\mathbf{P}}{3\epsilon_0}. \quad (2.29)$$

The macroscopic polarization $\mathbf{P}$ will be given by

$$\mathbf{P} = N\epsilon_0\chi_a\mathcal{E}_{\text{local}}, \quad (2.30)$$

where χ_a is the electric susceptibility per atom. χ_a is defined by:

$$\mathbf{p} = \epsilon_0\chi_a\mathcal{E}_{\text{local}}, \quad (2.31)$$

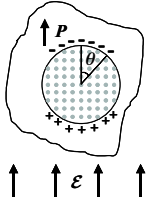


Fig. 2.8 Model used to calculate the local field by the Lorentz correction. An imaginary spherical surface drawn around a particular atom divides the medium into nearby dipoles and distant dipoles. The field at the centre of the sphere due to the nearby dipoles is summed exactly, while the field due to the distant dipoles is calculated by treating the material outside the sphere as a uniformly polarized dielectric.

$\mathbf{p}$ being the induced dipole moment per atom. This is analogous to the usual definition of the macroscopic susceptibility given in eqn A.1, except that it is now applied to individual atoms interacting with the local field. We can see from eqn 2.10 that χ_a is given by

$$\chi_a = \frac{e^2}{\epsilon_0 m_0} \frac{1}{(\omega_0^2 - \omega^2 - i\gamma\omega)}, \quad (2.32)$$

if there is just a single resonance. This is modified to

$$\chi_a = \frac{e^2}{\epsilon_0 m_0} \sum_j \frac{f_j}{(\omega_j^2 - \omega^2 - i\gamma_j\omega)}, \quad (2.33)$$

if there are multiple resonances (cf. eqn 2.24).

We can combine eqns 2.29 and 2.30 with eqns 2.11 and 2.13 by writing

$$\mathbf{P} = N\epsilon_0\chi_a \left(\mathcal{E} + \frac{\mathbf{P}}{3\epsilon_0} \right) = (\epsilon_r - 1)\epsilon_0\mathcal{E}. \quad (2.34)$$

We put all this together to find that:

$$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\chi_a}{3}. \quad (2.35)$$

This result is known as the **Clausius–Mossotti relationship**. The relationship works well in gases and liquids. It is also valid for those crystals in which the Lorentz correction given in eqn 2.29 gives an accurate account of the local field effects, namely cubic crystals.

2.3 The Kramers–Kronig relationships

The discussion of the dipole oscillator shows that the refractive index and the absorption coefficient are not independent parameters but are related to each other. This is a consequence of the fact that they are derived from the real and imaginary parts of a single parameter, namely the complex refractive index. If we invoke the law of causality (that an effect may not precede its cause) and apply complex number analysis, we can derive general relationships between the real and imaginary parts of the refractive index. These are known as the **Kramers–Kronig relationships** and may be stated as follows:

$$n(\omega) - 1 = \frac{2}{\pi} \mathcal{P} \int_0^\infty \frac{\omega' \kappa(\omega')}{\omega'^2 - \omega^2} d\omega' \quad (2.36)$$

$$\kappa(\omega) = -\frac{2}{\pi\omega} \mathcal{P} \int_0^\infty \frac{\omega'^2 [n(\omega') - 1]}{\omega'^2 - \omega^2} d\omega', \quad (2.37)$$

where $\mathcal{P}$ indicates that we take the principal part of the integral.

The Kramers–Kronig relationships allow us to calculate n from κ , and vice versa. This can be very useful in practice, because it would allow us, for example, to measure the frequency dependence of the optical

The derivation of the Kramers–Kronig relationships may be found, for example, in Dressel and Grüner (2002). The principal part of an integral that has a divergence at c within the integration range $a \rightarrow b$ is defined as:

$$\mathcal{P} \int_a^b f(x) dx = \lim_{\delta \rightarrow 0} \left(\int_a^{c-\delta} f(x) dx + \int_{c+\delta}^b f(x) dx \right).$$

absorption and then calculate the dispersion without needing to make a separate measurement of n .

As examples of the way the Kramers–Kronig relationships are used, we consider here two points raised in Section 1.4 of Chapter 1.

- (1) In the discussion of the data in Table 1.2 in Section 1.4.1, we noted that the refractive index of a semiconductor in its transparency range tends to increase with the band gap wavelength. Example 2.2 below shows that the refractive index at frequencies well below an absorption band centred at ω_0 has a contribution that varies as $1/\omega_0$. In the case of a semiconductor, we can put $\omega_0 \approx E_g/\hbar = 2\pi c/\lambda_g$, which then implies that the refractive index should increase linearly with λ_g . Figure 2.9 plots the refractive index data from Table 1.2 against the band gap wavelength. It is apparent that the relationship between n and λ_g is indeed approximately linear, which confirms the basic point that the higher frequency absorption in larger band gap materials gives a smaller contribution to the refractive index through the ω'^2 term in the denominator of eqn 2.36. The fact that the graph does not extrapolate to $n = 1$ at $\lambda_g = 0$ indicates that the narrow absorption band assumption of Example 2.2 should not be taken too far, and that there are other contributions to the refractive index that have not been considered here.

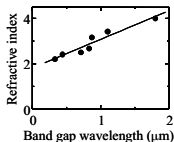


Fig. 2.9 Refractive index measured at 10 μm against band gap wavelength for several different semiconductors. The data are taken from Table 1.2.

- (2) In the discussion of the data in Table 1.4 in Section 1.4.2, we noted that the addition of compounds such as PbO to SiO₂ glass tends to reduce the frequency of the fundamental absorption edge and increase the refractive index in the visible spectral region. We can explain this by using a very simple model of the glass in which we assume that the material has an absorption band that runs from photon energy E_1 to E_2 . If we assume that the absorption has a constant value of α_0 throughout the band, a Kramers–Kronig analysis shows that the refractive index at a photon energy E below the absorption edge at E_1 is given by (see Exercise 2.12):

$$n(E) = 1 + \frac{c\hbar\alpha_0}{2\pi E} \ln \frac{(E_2 - E)(E_1 + E)}{(E_2 + E)(E_1 - E)}. \quad (2.38)$$

For pure SiO₂, the value of E_1 is about 10 eV, which corresponds to a wavelength of 120 nm. The additives reduce E_1 , which has the effect of increasing n through the relatively small $(E_1 - E)$ term in the denominator.

Example 2.2

A solid has a single absorption band of width γ centred at frequency ω_0 such that the extinction coefficient may be written in the form:

$$\begin{aligned} \kappa(\omega) &= \kappa_0, \quad \omega_0 - \gamma/2 \leq \omega \leq \omega_0 + \gamma/2, \\ &= 0 \quad \text{otherwise.} \end{aligned}$$

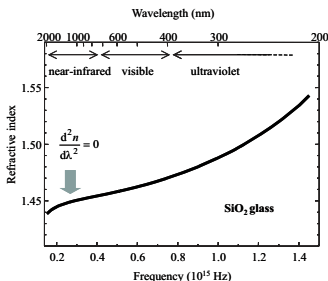


Fig. 2.10 Refractive index of SiO_2 glass in the near infrared, visible, and ultraviolet spectral regions. The thick arrow indicates the frequency at which the group velocity dispersion is zero. After Palik (1985).

Calculate the refractive index at low frequencies on the assumption that $\omega_0 \gg \gamma$.

Solution

We are asked to compute the refractive index at low frequencies, which means that we can assume $\omega_0 \gg \omega$, and therefore put $\omega = 0$ in the Kramers–Kronig relationships. On substituting into eqn 2.36 and using the fact that $\kappa(\omega)$ is zero at all frequencies apart from those within the absorption band, we can write:

$$n(0) = 1 + \frac{2}{\pi} \int_{\omega_0 - \gamma/2}^{\omega_0 + \gamma/2} \frac{\kappa_0}{\omega'} d\omega'.$$

Since $\omega_0 \gg \gamma$, we can take $\omega' = \text{constant} = \omega_0$ in the denominator of the integrand, and then obtain:

$$n(0) = 1 + \frac{2}{\pi} \frac{\kappa_0}{\omega_0} \times \gamma = 1 + \frac{2\gamma\kappa_0}{\pi\omega_0}.$$

2.4 Dispersion

Figure 2.10 plots the refractive index data from Fig. 2.7 in more detail. The data show that the refractive index increases with frequency in the near infrared and visible spectral regions. We have seen in Section 2.2.3 that this dispersion originates mainly from the interband absorption in the ultraviolet spectral region. At visible frequencies the absorption

from these transitions is negligible and the glass is transparent. However, the ultraviolet absorption still affects the refractive index through the extreme wings of the Lorentzian line. In the near infrared, the dispersion is also affected by the high-frequency wings of the vibrational absorption at lower frequency.

A material in which the refractive index increases with frequency is said to have **normal** dispersion, while one in which the contrary occurs is said to have **anomalous** dispersion. A number of empirical formulae to describe the normal dispersion of glasses have been developed over the years. (See Exercise 2.13.)

The dispersion of the refractive index of glasses such as silica can be used to separate different wavelengths of light with a prism, as shown in Fig. 2.11. The blue light is refracted more because of the higher index of refraction, and is therefore deviated through a larger angle by the prism. (See Exercise 2.14.) This effect is used in prism spectrometers.

One of the effects of dispersion is that light of different frequencies takes a different amount of time to propagate through a material. A pulse of light of duration t_p must necessarily contain a spread of frequencies given approximately by

$$\Delta\nu \approx \frac{1}{t_p} \quad (2.39)$$

in order to satisfy the ‘uncertainty principle’ $\Delta\nu\Delta t \sim 1$. Dispersion will therefore cause the pulse to broaden in time as it propagates through the medium, due to the different velocities of the frequency components of the pulse. This can become a serious problem when attempting to transmit very short pulses through a long length of an optical material, for example in a high speed optical fibre telecommunications system.

We mentioned in Section 2.2.3 that a pulse of light travels with the group velocity v_g . The important parameter for pulse spreading due to dispersion is therefore the **group velocity dispersion** (GVD). It is shown in Exercise 2.15 that the temporal broadening of a pulse due to group velocity dispersion is proportional to the second derivative of the refractive index with respect to the vacuum wavelength, and it is therefore useful to define a **material dispersion parameter** as follows:

$$D = -\frac{\lambda}{c} \frac{d^2n}{d\lambda^2}. \quad (2.40)$$

This is usually quoted in units of $\text{ps nm}^{-1} \text{km}^{-1}$, so that the temporal broadening $\Delta\tau$ of a pulse is given by:

$$\Delta\tau \text{ (ps)} = |D| \Delta\lambda L, \quad (2.41)$$

where $\Delta\lambda$ is the spectral width of the pulse measured in nm and L is the length of the medium in km.

The Lorentz model indicates that, within the normal dispersion region, $d^2n/d\lambda^2$ is negative for frequencies below an absorption line and positive above it. On applying this to the data in Fig. 2.10, we have a positive material dispersion parameter in the infrared due to the vibrational absorption and a negative one in the visible due to the interband

The use of the words ‘normal’ and ‘anomalous’ is somewhat misleading here. The dipole oscillator model shows us that all materials have anomalous dispersion at some frequencies. The phraseology was adopted before measurements of the refractive index had been made over a wide range of frequencies and the origin of dispersion had been properly understood.



Fig. 2.11 Separation of white light into different colours by dispersion in a glass prism.

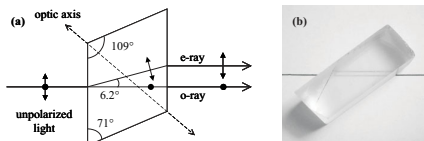


Fig. 2.12 (a) Double refraction in a calcite crystal. The shape of the crystal and the orientation of the optic axis is determined by the natural cleavage planes of calcite. An unpolarized incident light ray is split into two spatially separated, orthogonally polarized rays. The $\bullet$ symbol for the o-ray indicates that it is polarized with its field pointing out of the page. (b) An example of double refraction in a birefringent crystal. The single line underneath the crystal appears double, due to the separation of the o- and e-rays. Photo courtesy of S. Collins.

absorption in the ultraviolet. These two effects cancel at a wavelength in the near infrared, when the curvature of the graph of n against ω changes from negative to positive. This region of zero group velocity dispersion occurs around $1.3\ \mu\text{m}$ in silica optical fibres, and is identified in Fig. 2.10. Short pulses can be transmitted down the fibre with very small temporal broadening at this wavelength, and so it is one of the preferred wavelengths for optical fibre communication systems.

2.5 Optical anisotropy

For liquids and gases, it is reasonable to assume that the optical properties are isotropic, i.e. the same in all directions. This will also be the case for glasses and amorphous materials, which have no preferred physical axes. However, it will not, in general, be a valid assumption for crystals, which have well-defined axes arising from their structure. In this section we shall discuss the effects of anisotropy on optical materials, beginning with the natural anisotropy found in crystals, and then moving on to induced anisotropy caused by strain and external fields.

2.5.1 Natural anisotropy: birefringence

The atoms in a solid are locked into a crystalline lattice with well-defined axes. In general, we cannot assume that the optical properties along the different crystalline axes are equivalent. For example, the separation of the atoms might not be the same in all directions. This would lead to different vibrational frequencies, and hence a change in the refractive index between the relevant directions. Alternatively, the molecules locked into the lattice might preferentially absorb certain polarizations of light.

One of the most clear manifestations of optical anisotropy is the phenomenon of **birefringence**, which can be observed in transparent, anisotropic crystals. We can describe the properties of a birefringent

crystal by generalizing the relationship between the polarization and the applied electric field. If the electric field is applied along an arbitrary direction relative to the crystalline axes, we must write a tensor equation to relate $\mathbf{P}$ to $\mathbf{E}$:

$$\mathbf{P} = \epsilon_0 \boldsymbol{\chi} \mathbf{E} \quad (2.42)$$

where $\boldsymbol{\chi}$ represents the **susceptibility tensor**. Written explicitly in terms of the components, we have:

$$P_i = \epsilon_0 \sum_j \chi_{ij} E_j. \quad (2.43)$$

This can be conveniently expressed in terms of matrices as:

$$\begin{pmatrix} P_x \\ P_y \\ P_z \end{pmatrix} = \epsilon_0 \begin{pmatrix} \chi_{11} & \chi_{12} & \chi_{13} \\ \chi_{21} & \chi_{22} & \chi_{23} \\ \chi_{31} & \chi_{32} & \chi_{33} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}. \quad (2.44)$$

We can simplify the form of $\boldsymbol{\chi}$ by choosing the Cartesian coordinates x , y , and z to correspond to the principal axes of the crystal. In this case, the off-diagonal components are zero, and the susceptibility tensor takes the form:

$$\boldsymbol{\chi} = \begin{pmatrix} \chi_{11} & 0 & 0 \\ 0 & \chi_{22} & 0 \\ 0 & 0 & \chi_{33} \end{pmatrix}. \quad (2.45)$$

The relationships between the components are determined by the crystal symmetry.

- In cubic crystals, the x , y , and z axes are indistinguishable. They therefore have $\chi_{11} = \chi_{22} = \chi_{33}$, and their optical properties are isotropic.
- Crystals with tetragonal, hexagonal, or trigonal (rhombohedral) symmetry are called **uniaxial** crystals. These crystals possess a single **optic axis**, which is usually taken to be the z axis. In hexagonal crystals, for example, the optic axis is defined by the direction normal to the plane of the hexagons. The optical properties are the same along the x and y directions, but not along the z direction. This implies that $\chi_{11} = \chi_{22} \neq \chi_{33}$. Some examples of uniaxial crystals are listed in Table 2.1.
- Crystals with orthorhombic, monoclinic, or triclinic symmetry are called **biaxial** crystals. They have two optic axes, and all three diagonal components of the susceptibility tensor are different. Mica is an important example of a biaxial crystal.

One very striking demonstration of birefringence is the phenomenon of **double refraction**. In this effect an unpolarized light ray is separated into two rays which emerge displaced from each other, as shown in Fig. 2.12. These two rays are called ‘ordinary’ and ‘extraordinary’, and are orthogonally polarized to each other. It is apparent from Fig 2.12

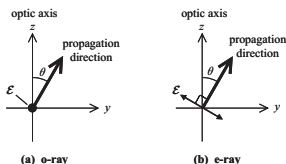
Equation 2.42 should be contrasted with the usual *scalar* relationship between $\mathbf{P}$ and $\mathbf{E}$ namely (cf. eqn A.1):

$$\mathbf{P} = \epsilon_0 \chi \mathbf{E},$$

which only applies to isotropic materials.

Crystals with cubic symmetry are only isotropic as regards their *linear* optical properties. We shall see in Chapter 11 that cubic crystals can actually have anisotropic *nonlinear* optical properties. This is possible because the nonlinear properties are described by *second-rank* tensors (e.g. $\chi_{ijk}^{(2)}$), rather than the *first-rank* tensors (e.g. χ_{ij}) used for the linear optical properties. See Nye (1985) for further details.

Fig. 2.13 Electric field vector of a ray propagating in a uniaxial crystal with its optic axis along the z direction. The propagation direction is determined by the direction of the Poynting vector. The ray makes an angle of θ with respect to the optic axis. The x and y axes are chosen so that the beam is propagating in the y, z plane. The polarization can be resolved into: (a) a component along the x axis and (b) a component at an angle of $90^\circ - \theta$ to the optic axis. (a) is the o-ray and (b) is the e-ray.



that the e-ray does not obey Snell's law of refraction, which is why it is called 'extraordinary'.

The phenomenon of double refraction can be explained by assuming that the crystal has different refractive indices for the orthogonal polarizations of the ordinary and extraordinary rays. These two refractive indices are usually labelled n_o and n_e respectively. Consider the propagation of a beam of unpolarized light that enters a uniaxial crystal at an angle θ to the optic axis, which is taken to lie along the z axis. The optical properties are isotropic in the x, y plane, and so we can choose the axes so that the beam is propagating in the y, z plane without loss of generality, as shown in Fig. 2.13. This allows us to split the polarization of the light into two orthogonal components, one of which is polarized along the x axis, and the other polarized at an angle of $(90^\circ - \theta)$ to the optic axis. The former is the o(rdinary)-ray, and the latter is the e(xtraordinary)-ray. Now the refractive index will be different for light which is polarized along the z axis or in the x, y plane. Therefore the o-ray, which has no polarization component along the z axis, experiences a different refractive index to the e-ray, which has a component along z . The two rays will thus be refracted differently: hence double refraction. On the other hand, if the beam propagates along the optic axis so that $\theta = 0$, the $\mathcal{E}$ -vector of the light will always fall in the x, y plane. In this case, no double refraction will be observed because the x and y directions are equivalent.

Double refraction was first observed in natural uniaxial crystals such as calcite ('Iceland Spar'). Table 2.1 lists the refractive indices for the o- and e-rays of calcite, together with those of several other uniaxial crystals. The birefringent crystals are classified as being either positive or negative depending on whether n_e is larger or smaller than n_o .

Uniaxial birefringent crystals find widespread application in making optical components to control the polarization state of light. Figure 2.14 illustrates the operating principle of a **Glan-Foucault polarizing prism**. The polarizer consists of two identical birefringent prisms mounted with an air gap between them, with their optic axes in the plane of the front surface. Unpolarized light at the input surface of the prism is split into the o-ray and e-ray, which are then incident at the intermediate

Table 2.1 Refractive indices of some common uniaxial crystals at 589.3 nm. Data from Driscoll & Vaughan (1978).

Crystal	Chemical structure	Symmetry class	Type	n_o	n_e
Ice	H ₂ O	trigonal	positive	1.309	1.313
Quartz	SiO ₂	trigonal	positive	1.544	1.553
Beryl	Be ₃ Al ₂ (SiO ₃) ₆	hexagonal	negative	1.581	1.575
Sodium nitrate	NaNO ₃	trigonal	negative	1.584	1.336
Calcite	CaCO ₃	trigonal	negative	1.658	1.486
Tourmaline	complex silicate	trigonal	negative	1.669	1.638
Sapphire	Al ₂ O ₃	trigonal	negative	1.768	1.760
Zircon	ZrSiO ₄	tetragonal	positive	1.923	1.968
Rutile	TiO ₂	tetragonal	positive	2.616	2.903

air interface. For the case of positive uniaxial crystals, the apex angle θ of the prisms is chosen so that the o-ray suffers total internal reflection, but not the e-ray. (See Exercise 2.17.) The light that emerges at the output of the polarizer therefore only consists of the e-ray, and is linearly polarized. The Glan–Foucault prism therefore turns unpolarized light into linearly polarized light. The **Glan–Thompson polarizing prism** is a variation of the Glan–Foucault prism, with optical cement in the gap between the two birefringent prisms. This improves the acceptance angle of the polarizer (i.e. the tolerance to the angle of incidence of the input beam) at the expense of reducing the optical damage threshold. This makes the Glan–Thompson prism more useful for general purpose optics, but less useful when working with high power lasers.

Figure 2.15 illustrates another important optical component that exploits birefringence, namely the **retarder plate**. The retarder is made by cutting a uniaxial birefringent crystal so that the optic axis lies in the plane of the input surface of the plate. Figure 2.15 shows a retarder plate with a linearly polarized input beam. As the beam propagates through the crystal, it can be resolved into an o-ray and r-ray, as shown in Fig. 2.15(b). The two rays experience different refractive indices, and therefore propagate with different phase velocities. This introduces a relative phase difference (or ‘retardation’) between the o- and e-rays. The magnitude $\Delta\phi$ of the phase difference is given by:

$$\Delta\phi = \frac{2\pi|n_o - n_e|d}{\lambda} = \frac{2\pi|\Delta n|d}{\lambda}, \quad (2.46)$$

where d is the thickness of the retarder plate, λ is the vacuum wavelength of the light, and Δn is the difference between the ordinary and extraordinary refractive indices. The thickness of the plate is usually chosen so that $\Delta\phi$ equals either $\pi/2$ or π . For the case of $\Delta\phi = \pi/2$, the phase difference is equivalent to a quarter of a wave, and so the retarder is called a **quarter-wave plate**. For similar reasons, a plate with $\Delta\phi = \pi$ is called a **half-wave plate**. A quarter-wave plate will turn linearly-polarized light into circularly-polarized light, and vice versa, while a half-wave plate will rotate the polarization of linearly polarized light.

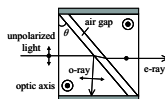
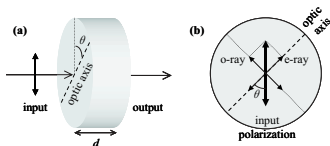
**Fig. 2.14** Glan–Foucault polarizing prism. The optic axes of the crystals point vertically out of the page. In the case shown here, the crystals are assumed to have positive birefringence. The role of the o- and e-rays are reversed if the crystals have negative birefringence.

Fig. 2.15 (a) A birefringent optical retarder plate with linear input polarization. The optic axis of the birefringent crystal lies in the plane of the input surface. (b) Input surface of the retarder, showing the resolution of the input polarization into the o- and e-rays. θ is the angle between the input polarization and the optic axis, and d is the thickness of the plate.



(See Exercise 2.18.)

Further discussion of the detailed effects of birefringence may be found in most optics textbooks. The main purpose of discussing birefringence here is to illustrate the phenomenon of optical anisotropy and make the point that it arises from the underlying symmetry of the crystal structure. This is a very standard example of an optical effect that occurs in crystalline solids and is not found in gases or liquids.

Example 2.3

The optic axis of a uniaxial crystal lies along the z axis. The refractive index for light polarized in the z direction is n_e , while that for light polarized in the x, y plane is n_o . Write down the dielectric constant tensor defined through the tensor relationship

$$\mathbf{D} = \epsilon_0 \epsilon_r \mathbf{E}.$$

Solution

We make use of eqns 2.11 and 2.42 to write:

$$\begin{aligned} \mathbf{D} &= \epsilon_0 \mathbf{E} + \mathbf{P} \\ &= \epsilon_0 \mathbf{E} + \epsilon_0 \chi \mathbf{E} \\ &= \epsilon_0 (1 + \chi) \mathbf{E} \equiv \epsilon_0 \epsilon_r \mathbf{E}. \end{aligned} \quad (2.47)$$

Hence we see that:

$$\epsilon_r = 1 + \chi. \quad (2.48)$$

The susceptibility tensor is given by eqn 2.45, and hence the dielectric constant tensor will take the form:

$$\epsilon_r = \begin{pmatrix} 1 + \chi_{11} & 0 & 0 \\ 0 & 1 + \chi_{22} & 0 \\ 0 & 0 & 1 + \chi_{33} \end{pmatrix}. \quad (2.49)$$

In a uniaxial crystal with the optic axis along the z direction, we must have $\chi_{11} = \chi_{22} \neq \chi_{33}$.

We now further assume that the crystal is transparent, so that the dielectric constant is just equal to the square of the refractive index (cf. eqns 1.27 and 1.28 with $\kappa = 0$). If we had a linearly polarized light beam with the electric field directed along the x or y directions, we would measure a refractive index of n_o . This tells us that

$$1 + \chi_{11} = 1 + \chi_{22} = n_o^2.$$

On the other hand, if $\mathcal{E}$ is along the z axis, we would measure a refractive index of n_e , which implies that

$$1 + \chi_{33} = n_e^2.$$

Therefore the dielectric constant tensor must be:

$$\epsilon_r = \begin{pmatrix} n_o^2 & 0 & 0 \\ 0 & n_o^2 & 0 \\ 0 & 0 & n_e^2 \end{pmatrix}. \quad (2.50)$$

Example 2.4

Calculate the thickness of a quartz half-wave plate designed for a wavelength of 589 nm.

Solution

The thickness can be calculated by substituting the refractive index data from Table 2.1 into eqn 2.46. With $n_o = 1.544$ and $n_e = 1.553$, we have $|\Delta n| = 0.009$. In a half-wave plate we require $\Delta\phi = \pi$, and so we can find d by solving:

$$\Delta\phi = \pi = \frac{2\pi|\Delta n|d}{\lambda}.$$

This gives $d = \lambda/2|\Delta n| = 0.033$ mm.

A quartz plate of thickness 0.033 mm would be very fragile, and so optical companies frequently design their wave plates to have a retardation of $(2\pi m + \Delta\phi)$, where m is an integer. This makes no difference at the design wavelength, and allows more practical thicknesses to be used.

2.5.2 Induced optical anisotropy

Isotropic materials such as liquids, gases, and glasses are not birefringent. However, the application of external perturbations can break the symmetry, thereby producing birefringence. This gives rise to a range of induced optical phenomena associated with strain and electric fields. Note that the application of a magnetic field induces optical activity rather than birefringence, and so it must be treated differently. (See Section 2.6 below.)

The most obvious way to break the symmetry of an isotropic medium is to compress it in one direction. The resulting strain-induced birefringence is called the **photo-elastic effect**. The effect can readily be observed by placing a piece of stressed glass between crossed polarizers.

Since an electromagnetic wave consists of an oscillating transverse electric field, a high-intensity light beam can produce self-induced birefringence through the Kerr effect. This is an example of a nonlinear optical effect, and will be discussed further in Chapter 11.

Table 2.2 Kerr constants of representative substances. In the case of oxide glasses, the Kerr constant generally increases with the PbO content. Data from Kaye & Laby (1986) and Hoffmann (1995).

Substance	K (m V^{-2})
Nitrobenzene	4.4×10^{-12}
CS_2	3.6×10^{-14}
Water	5.2×10^{-14}
Oxide glass	$0.1 - 3 \times 10^{-14}$
Chalcogenide glass (As_2O_3)	8.7×10^{-14}

In the absence of strain, the glass should have no effect on the polarization of the light, and so there should be no transmission through the second polarizer. However, if stress is present, the polarization vector will be altered, and some light will be transmitted. This method is in fact used to detect strain in glasses and other isotropic optical materials.

Birefringence can also be induced in an isotropic material by applying an electric field to break the symmetry. This effect was first discovered by Kerr in the nineteenth century, and is therefore known as the **Kerr effect**. Kerr discovered that an isotropic medium behaves as a uniaxial crystal when an electric field is applied in the direction transverse to the light direction. The optic axis is parallel to the field, and the induced birefringence is given by:

$$\Delta n = \lambda K \mathcal{E}^2, \quad (2.51)$$

where λ is the vacuum wavelength, K is the Kerr constant, and $\mathcal{E}$ in the field strength. Since the birefringence is proportional to the square of the field, the Kerr effect is alternatively known as the **quadratic electro-optic effect**. Table 2.2 lists some representative values of the Kerr constant.

The quadratic field dependence of the Kerr effect can be understood in simple terms as follows: the first power of the field breaks the symmetry, and the second power induces the refractive index change. This contrasts with the **Pockels effect** which is observed in anisotropic crystals, where symmetry breaking is not required. Therefore, a refractive index change proportional to the field automatically produces induced birefringence, giving rise to a **linear electro-optic effect**.

In addition to the different functional dependence on the field strength, there are a number of other important differences between the linear and quadratic electro-optic effects.

- (1) The Kerr effect can, in principle, be observed in any medium, but the Pockels effect is only observed in anisotropic crystals that lack inversion symmetry.
- (2) The Kerr effect is only observed with transverse fields, but the Pockels effect can also be observed with longitudinal fields.
- (3) Since no symmetry breaking is required for the Pockels effect, the fields required to induce a particular value of Δn are smaller than for the Kerr effect, so that the Kerr effect is usually negligible in an anisotropic medium that exhibits the Pockels effect. The Kerr effect is normally only studied in isotropic media such as liquids, gases, and glasses, and requires large fields to observe significant effects. (See Exercise 2.21.)

Further discussion of the linear electro-optic effect and Kerr effect is postponed to Sections 11.3.4 and 11.4.3 in Chapter 11.

2.6 Optical chirality

Objects that do not possess reflection symmetry appear reversed when observed in a mirror. Left- and right-handed gloves are obvious examples, as are springs, corkscrews and helices. Such objects are said to possess **chirality** (i.e. handedness). The fact that an optical medium possesses chirality implies that its response to left- and right-circular light will be different. We can therefore quantify optical chirality by the difference of the refractive indices experienced by left- and right-circular light. A difference in the real part of the complex refractive index gives rise to **optical activity**, while a difference in the imaginary part causes **circular dichroism**.

Optical activity is observed in transparent, chiral materials, and causes the direction of linear polarization to rotate as the light propagates through the medium. The rotation can be in either direction, which leads to a sub-classification of the material as being either **dextro-rotatory** or **laevo-rotatory**, depending on whether the rotation is in a clockwise or anti-clockwise sense when looking along the beam from the source. Since optical activity arises from a difference in the refractive index for circularly polarized light, it is sometimes called **circular birefringence**. This should not be confused with the birefringence discussed in Section 2.5.1, which arises from anisotropy, and is quantified by the difference in the refractive indices for orthogonal *linear* polarizations.

The rotation angle θ of the polarization in an optically-active medium of thickness d is given by (see Exercise 2.22):

$$\theta = \frac{\pi d}{\lambda} (n_R - n_L), \quad (2.52)$$

where n_R (n_L) is the refractive index for right- (left-) circular light, and λ is the vacuum wavelength. A dextro-rotatory medium has $n_R < n_L$, and vice versa for a laevo-rotatory one.

Circular dichroism is observed in absorbing chiral materials. The chirality manifests itself as a difference in the imaginary part of the refractive index for left- and right-circular light. The word ‘dichroic’ means ‘two-coloured’, and the term dichroism is applied to any optical phenomenon that affects two colours differently. In the case of circular dichroism, the absorption coefficient is sensitive to the direction to the circular polarization, which implies that if the absorption bands lie in the visible spectral region, the colour will be different when viewed with left- or right-circular light.

Chirality can arise either from the crystalline structure, or from the molecules that comprise the medium. We concentrate here on the former effect, of which crystalline quartz (SiO_2) is a good example, with both dextro-rotatory and laevo-rotatory forms existing in nature. Quartz is a uniaxial birefringent crystal, and the optical activity is best observed when the light propagates along the optic axis. In this situation, linearly polarized light experiences a refractive index of n_o irrespective of its orientation, which means that it emerges unaffected in a non-chiral

The names ‘dextro-rotatory’ and ‘laevo-rotatory’ are derived from the Latin words *dexter* and *laevus* which mean ‘right’ and ‘left’ respectively. Dextro-rotatory and laevo-rotatory materials are alternatively said to have positive or negative chirality, respectively.

The study of the chirality of molecules is very important in chemistry and biology. Since the chirality arises from the molecules themselves, it can be observed even in isotropic media such as liquids. Sugar solution is a well-known example, so much so that dextrose and laevulose (commonly known as fructose) get their names from their dextro-rotatory and laevo-rotatory properties respectively.

Table 2.3 Verdet constant of representative substances. Since V varies with λ , the wavelength of the measurement must be specified. Data from Kaye & Laby (1986).

Substance	Wavelength (nm)	V (radian $\text{T}^{-1} \text{ m}^{-1}$)
Fused silica (SiO_2)	546.1	5.0
Hard crown glass	589.3	5.5
Light flint glass	589.3	9.0
Dense flint glass	589.3	11.2
Rock salt (NaCl)	670.8	7.1
Fluorite (CaF_2)	589.3	7.1
Water	589.3	3.8
CS_2	589.3	12.2

uniaxial crystal such as calcite. However, since the unit cell of quartz is chiral, linearly polarized light emerges rotated even when propagating along the optic axis.

The optical activity of crystalline quartz should be contrasted with the absence of optical activity in fused silica (SiO_2 glass). This shows that the optical activity arises from the crystal structure rather than the molecules. The unit cell of quartz belongs to the trigonal crystal class labelled 32, which has no mirror symmetry, and is therefore chiral. By contrast, calcite has the $\bar{3}2/m$ structure, where the 'm' symbol implies mirror symmetry in the unit cell, and therefore a lack of chirality.

Optical chirality can be induced in non-chiral materials by applying a magnetic field, thereby giving rise to a number of **magneto-optical** phenomena. In the case of transparent materials, the field induces optical activity and the phenomenon is called either the **Faraday effect** or the **magneto-optical Kerr effect**, depending on whether the rotation of the polarization is observed in transmission or reflection respectively. If the medium is absorbing, the field can induce circular dichroism, in which case we have **magnetic circular dichroism**. All of these magneto-optical effects ultimately have their origin in the Zeeman effect. (See Exercise 2.23.)

Figure 2.16 illustrates the Faraday effect. The field is applied along the axis of an optical medium and this causes a rotation of linearly polarized light. The rotation angle θ is related to the field strength B by:

$$\theta = V B d, \quad (2.53)$$

where V is the **Verdet coefficient** of the medium and d is its thickness. Table 2.3 lists the values of the Verdet constant for a number of materials. In general, the Verdet coefficients are small in the visible spectral region, and decrease strongly as the detuning from the ultraviolet absorption bands increases. In practice, this means that substantial thicknesses have to be used to obtain significant rotations at the field strengths that are readily available from permanent magnets. (See Exercise 2.24.)

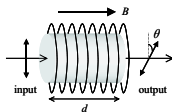


Fig. 2.16 The Faraday effect. A magnetic field B applied along the axis of an optical medium induces optical activity and causes a rotation of linearly polarized light by an angle θ .

Chapter summary

- The classical model of a solid treats the atoms and molecules as oscillating electric dipoles with characteristic resonant frequencies. The resonances due to the bound electrons occur in the near infrared, visible, and ultraviolet spectral regions (10^{14} – 10^{15} Hz), while those associated with vibrations occur in the infrared (10^{12} – 10^{13} Hz). Free electrons can be treated in the dipole oscillator model by assuming that the natural resonant frequency is zero.
- The medium absorbs light when the frequency coincides with one of its resonant frequencies. In non-resonant conditions, the medium is transparent, but the velocity of light is reduced by the phase lag due to multiple coherent elastic scattering.
- The absorption coefficient of an individual dipole oscillator has a Lorentzian line shape (cf. eqn 2.21). The spectral width of the absorption line is equal to the damping constant γ . The peak absorption is proportional to $1/\gamma$.
- The refractive index of a dipole oscillator increases as the frequency approaches the resonant frequency, then drops sharply in the absorbing region, and then increases again at higher frequencies. The off-resonant refractive index decreases each time we increase the frequency through an absorption line.
- The relative dielectric constant of a medium with multiple resonant frequencies is given by eqn 2.24. The refractive index and absorption coefficient can be calculated from the real and imaginary parts of ϵ_r .
- The dipole oscillator model demonstrates that the absorption and refraction of an optical medium are fundamentally related to each other. This interrelationship is made explicit through the Kramers–Kronig formulae.
- Dispersion in the refractive index originates from the wings of the resonances at transition frequencies. The dispersion is called normal when the refractive index increases with frequency. Group velocity dispersion causes temporal broadening of short pulses.
- Optical anisotropy leads to birefringence. The anisotropy is described through the electric susceptibility tensor or the dielectric constant tensor. Anisotropy can be induced in isotropic media by strain or electric fields, giving rise respectively to the photo-elastic and electro-optic effects.
- Structural chirality causes optical activity and circular dichroism. The chirality can be induced by a magnetic field, which gives rise to magneto-optical effects.

Further reading

The subject matter of this chapter is covered, to a greater or lesser extent, in most electromagnetism and optics textbooks. See, for example: Bleaney and Bleaney (1976), Born and Wolf (1999), Hecht (2001), Smith, King, and Wilkins (2007), or Klein and Furtak (1986).

An excellent collection of optical data on a wide range of solid-state materials can be found in Palik (1985). A

detailed discussion on the origin of the dispersion in silica glass and its relationship to ultraviolet and infrared absorption may be found in Smith et al. (2004).

For a fuller description of birefringence and optical activity, see: Hecht (2001), Smith, King, and Wilkins (2007), Born and Wolf (1999) or Klein and Furtak (1986).

Exercises

- (2.1) Write down the equations of motion for the frictionless displacements x_1 and x_2 of two masses, m_1 and m_2 , connected together by a light spring with a spring constant K_s . Hence show that the angular frequency for small oscillations is equal to $(K_s/\mu)^{1/2}$ where $\mu^{-1} = m_1^{-1} + m_2^{-1}$.
- (2.2) A damped oscillator with mass m , natural angular frequency ω_0 , and damping constant γ is being driven by a force of amplitude F_0 and angular frequency ω . The equation of motion for the displacement x of the oscillator is:

$$m \frac{d^2 x}{dt^2} + m\gamma \frac{dx}{dt} + m\omega_0^2 x = F_0 \cos \omega t.$$

What is the phase of x relative to the phase of the driving force?

- (2.3) A sapphire crystal doped with titanium absorbs strongly around 500 nm. Calculate the difference in the refractive index of the doped crystal above and below the 500 nm absorption band if the density of absorbing atoms is $1 \times 10^{25} \text{ m}^{-3}$. The refractive index of undoped sapphire is 1.77.
- (2.4) The laser crystal $\text{Ni}^{2+}:\text{MgF}_2$ has a broad absorption band in the blue which peaks at 405 nm and has a full width at half maximum of $8.2 \times 10^{13} \text{ Hz}$. The oscillator strength of the transition is 9×10^{-5} . Estimate the maximum absorption coefficient in a crystal with $2 \times 10^{26} \text{ m}^{-3}$ absorbing atoms per unit volume. The refractive index of the crystal is 1.39.
- (2.5) Show that the absorption coefficient of a Lorentz oscillator at the line centre does not depend on the value of ω_0 .

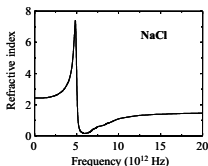


Fig. 2.17 Infrared refractive index of NaCl. After Palik (1985).

- (2.6) Figure 2.17 shows the refractive index of NaCl in the infrared spectral region. The data can be modelled approximately by assuming that the resonance feature is caused by the vibrations of the completely ionic Na^+Cl^- molecules. The atomic weights of sodium and chlorine are 23 and 35.5 respectively. Use the data to estimate:
- The static dielectric constant of NaCl.
 - The natural oscillation frequency of the vibrations.
 - The restoring force for a unit displacement of the oscillator.
 - The density of NaCl molecules per unit volume.
 - The damping constant γ for the vibrations.
 - The peak absorption coefficient.
- (2.7) Derive both the relationships between the group

velocity and the phase velocity given in eqn 2.26.

- (2.8)* Consider a simple Lorentz oscillator with a single undamped resonance. The dielectric constant will be given by eqn 2.14 with χ and γ both zero. This gives:

$$\epsilon_r(\omega) = 1 + \frac{Ne^2}{\epsilon_0 m_0} \frac{1}{(\omega_0^2 - \omega^2)}.$$

Prove that the group velocity is always less than c .

- (2.9)* Consider a dielectric sample placed in a uniform electric field pointing in the z direction as shown in Fig. 2.8. Assume that the atoms are arranged on a cubic lattice and the dipoles are all pointing along the external field direction.

(a) Let us first consider the field generated by the dipoles within the spherical surface. By using the standard formula for the electric field generated by an electric dipole, show that the field at the centre of the sphere is given by

$$\mathcal{E}_{\text{sphere}} = \frac{1}{4\pi\epsilon_0} \sum_j p_j \frac{3z_j^2 - r_j^2}{r_j^5},$$

where the summation runs over all the dipoles within the surface except the one at the centre, and p_j is the dipole moment of the atom at the j th lattice site.

(b) Show that $\mathcal{E}_{\text{sphere}} = 0$ in a homogenous medium where all the values of p_j are identical.

(c) Now consider the uniformly polarized dielectric material outside the spherical hole. Let $\mathbf{P}$ be the macroscopic polarization of the medium, which is assumed to be parallel to the external field. Show that the surface charge density on the sphere at an angle θ from the z axis is equal to $-P \cos \theta$. Hence show that the material outside the spherical surface generates a field at the centre of the sphere equal to $-\mathbf{P}/3\epsilon_0$.

- (2.10) Under what conditions does the Clausius–Mossotti relationship given by eqn 2.35 reduce to the usual relationship between the dielectric constant and the electric susceptibility given in eqn A.4?

- (2.11) The relative dielectric constant of N_2 gas at standard temperature and pressure is 1.000588. Calculate χ_a for the N_2 molecule. Show that the electric field strength required to generate a dipole equivalent to displacing the electron by $1 \text{ \AA}$ (10^{-10} m) is of a similar magnitude to the electric field between a proton and an electron separated by the

same distance.

- (2.12) A certain material has a single absorption band which runs from photon energy E_1 to E_2 . On the assumption that the absorption coefficient has a constant value of α_0 within the band, and is zero elsewhere, use the Kramers–Kronig relationship to show that the refractive index for a photon energy E below E_1 is given by:

$$n(E) = 1 + \frac{c\hbar\alpha_0}{2\pi E} \ln \frac{(E_2 - E)(E_1 + E)}{(E_2 + E)(E_1 - E)}.$$

- (2.13) (a) Sellmeier derived the following equation for the wavelength dependence of the refractive index in 1871:

$$n^2 = 1 + \sum_j \frac{A_j \lambda^2}{(\lambda^2 - \lambda_j^2)}.$$

Show that this equation is equivalent to eqn 2.24 in regions of transparency far from any absorption lines. State the values of A_j and λ_j .

(b) Assume that the dispersion is dominated by the closest resonance, so that we only need to include one term (say the one with $j = 1$) in the summation of Sellmeier's equation. Assume that λ_1^2/λ^2 is small, and expand Sellmeier's equation to derive the earlier dispersion formula determined empirically by Cauchy:

$$n = C_1 + \frac{C_2}{\lambda^2} + \frac{C_3}{\lambda^4} + \dots$$

State the values of C_1 , C_2 , and C_3 in terms of A_1 and λ_1 .

- (2.14) The refractive index of crown glass is 1.5553 at 402.6 nm and 1.5352 at 706.5 nm.

(a) Determine the coefficients C_1 and C_2 in Cauchy's formula given in the previous exercise, on the assumption that the term in C_3 is negligible.

(b) Estimate the refractive index for blue light at 450 nm and for red light at 650 nm.

(c) White light strikes a crown glass prism with an apex angle of 60° , as shown in Fig. 2.11. The angle of incidence with the first surface is 45° . Calculate the difference in the angle between the light at 450 nm and 650 nm at the exit surface of the prism.

- (2.15) Show that the temporal broadening of a short pulse by a dispersive medium of length L is given approximately by:

$$\Delta\tau = L \left| \frac{\lambda}{c} \frac{d^2 n}{d\lambda^2} \right| \Delta\lambda,$$

*Exercises marked with an asterisk are more difficult.

where λ is the vacuum wavelength and $\Delta\lambda$ is the spectral width of the pulse. Estimate $\Delta\tau$ for a laser pulse with a temporal width of 10 ps in 1 km of optical fibre at 1550 nm, where $|(\lambda/c)d^2n/d\lambda^2| = 17 \text{ ps km}^{-1} \text{ nm}^{-1}$.

- (2.16) Consider the propagation of a wave with polarization vector components (x, y, z) , where $x^2 + y^2 + z^2 = 1$, in a birefringent medium. The dielectric constant experienced by the wave is conveniently described by the index ellipsoid:¹

$$\frac{x^2}{\epsilon_{11}/\epsilon_0} + \frac{y^2}{\epsilon_{22}/\epsilon_0} + \frac{z^2}{\epsilon_{33}/\epsilon_0} = 1,$$

where the ϵ_{ij} are the components of the dielectric constant tensor defined in eqn 2.49. Use the index ellipsoid to show that the refractive index for the e-ray propagating at an angle θ to the optic axis of a uniaxial crystal as shown in Fig. 2.13(b) is given by:

$$\frac{1}{n(\theta)^2} = \frac{\sin^2 \theta}{n_e^2} + \frac{\cos^2 \theta}{n_o^2},$$

where n_e and n_o are defined in Example 2.3.

- (2.17) By using the refractive index data given in Table 2.1, calculate the range of apex angles that will lead to selective total internal reflection of the o-ray in a Glan-Foucault polarizing prism made from calcite.

- (2.18) (a) Consider a half-wave plate with linearly polarized light at the input surface, as shown in Fig. 2.15. Show that the wave plate rotates the polarization of the light by an angle 2θ , where θ is the angle between the input polarization and the optic axis.

(b) Show that a quarter-wave plate turns a linearly polarized input beam into circularly-polarized light, and vice versa, if the angle θ is chosen to be 45° .

(c) Describe the output of the quarter-wave plate when $\theta \neq 45^\circ$.

- (2.19) A uniaxial birefringent crystal made from quartz has $n_o = 1.5443$ and $n_e = 1.5534$. A retardation plate is made by cutting the crystal so that the optic axis is parallel to the surfaces of the plate, as shown in Fig. 2.15. Calculate the thickness of the crystal that must be chosen so that it behaves as a quarter-wave plate at 500 nm.

- (2.20) Look up the crystal structures of the following materials to determine whether they are birefringent or not: (a) NaCl, (b) diamond, (c) graphite

(in the infrared, where it transmits), (d) ZnS (wurtzite structure), (e) ZnS (zinc-blende structure), (f) solid argon at 4 K, (g) sulphur. Specify which, if any, of the birefringent materials are biaxial.

- (2.21) (a) A Kerr cell consists of a Kerr medium with contacts attached so that an electric field can be applied. Show that the field strength required to produce a birefringent phase shift equivalent to half a wavelength is given by:

$$\mathcal{E}_{\lambda/2} = 1/\sqrt{2Kd},$$

where K is the Kerr constant, and d is the length of the medium.

(b) Calculate the voltage required to produce a field of $\mathcal{E}_{\lambda/2}$ in a Kerr cell of length 2 cm made from chalcogenide glass with a Kerr constant of $8.7 \times 10^{-14} \text{ m V}^{-2}$, if the lateral width across which the voltage is dropped is 5 mm.

- (2.22) (a) A transparent chiral medium of thickness d has a refractive index of n_L and n_R for left- and right-circular light respectively for light of vacuum wavelength λ . By considering linearly-polarized light as a superposition of left- and right-circular light, show that the medium rotates the linear polarization by an angle θ given by

$$\theta = \frac{\pi d}{\lambda} (n_R - n_L).$$

(b) The value of $|n_R - n_L|$ for crystalline quartz is 7.1×10^{-5} at 589 nm. Calculate the rotatory power of quartz (defined as θ/d) at this wavelength in units of $^\circ/\text{mm}$.

- (2.23) Consider an optical medium with a single Lorentzian resonance line at angular frequency ω_0 of width γ . When a magnetic field of strength B is applied, the transition energy shifts by $\pm \mu_B B$ according to the normal Zeeman effect. For light propagating parallel to the field, the absorption frequency increases to $\omega_0 + \mu_B B/\hbar$ for σ^+ circular polarization, and decreases to $\omega_0 - \mu_B B/\hbar$ for σ^- polarization. Magneto-optical phenomena can be explained by considering the difference in the complex refractive index, $(\tilde{n}_+ - \tilde{n}_-)$, for σ^+ and σ^- light.

(a) By considering the real part of $(\tilde{n}_+ - \tilde{n}_-)$, sketch the frequency dependence of the Faraday

¹The use of the index ellipsoid can be justified by considering the direction of the energy flow through the crystal: see Born and Wolf (1999).

rotation.

(b) By considering the imaginary part of $(\tilde{n}_+ - \tilde{n}_-)$, sketch the frequency dependence of the magnetic circular dichroism.

(2.24) In an optical isolator, the Faraday effect is used

to rotate the plane of linearly polarized light by 45° . Given that Verdet coefficient of flint glass is $9.0 \text{ radian T}^{-1} \text{ m}^{-1}$, calculate the thickness of glass required in an isolator with a permanent magnet of strength 0.5 T .